Adrian Cottam, Ph.D.



Assistant Research Professor, Auburn University Transportation Research Institute (AUTRI)

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Education

Ph.D. in Transportation Engineering (Minor in Management Information Systems) — May 2019 – Aug. 2023
Civil and Architectural Engineering and Mechanics, University of Arizona, Tucson, Arizona

Graduate Certificate in College Teaching — May 2019 – Dec. 2021
Office of Instruction and Assessment, University of Arizona, Tucson, Arizona

M.E. in Transportation Engineering — Aug. 2018 - May 2019
Civil and Architectural Engineering and Mechanics, University of Arizona, Tucson, Arizona

B.S. in Civil Engineering — Aug. 2016 - May 2018
Civil and Architectural Engineering and Mechanics, University of Arizona, Tucson, Arizona

A.S. in Civil Engineering – ***Highest Honors*** — Jun. 2013 - May 2016
Pima Community College, Tucson, Arizona

Research Interests

- **Intelligent Transportation Systems**
  - **Traffic sensor data collection and analysis**
  - **Performance measurement with emerging data sources**
- **Transportation Big Data Analyses**
  - **Freeway operations**
  - **Machine learning applications to transportation**
  - **Transportation data fusion**
  - **Large scale performance measurement**
- **Information Systems**
  - **Traffic data visualization dashboard development**
  - **Traffic database design and optimization**
- **Simulation**
  - **Advanced microsimulation analyses and calibration**
  - **Digital twin creation**
  - **Driving simulator systems development and studies**
- **Multi-Modal Transportation**
  - **Emerging modes (E-scooters, bike-sharing, sidewalk autonomous delivery robots)**
  - **Intermodal interactions**

Awards and Honors

- **National Institute for Transportation and Communities (NITC) Dissertation Fellowship,** Feb. 2023 – Aug. 2023
- **Dwight David Eisenhower Transportation Fellowship Program (DDETFP) Fellow,** Dec. 2021 – Oct. 2022.
- **Government Assistance in Areas of National Need (GAANN) Fellow,** Feb. 2019 – Spring 2022.
- **Lyon Family CEEM Scholarship,** Aug. 2020.
- **National Institute for Transportation and Communities (NITC) Student Scholar Award**. Aug. 2018.
- **National Institute for Transportation and Communities (NITC) Student Scholar Award**. Mar. 2018.
- **Southern Arizona Institute of Transportation Engineers (SAITE) Hank Warner Memorial Undergraduate Scholarship**, Mar. 2018.
- **Dwight Seiferlein Memorial Scholarship,** Sep. 2016.
- **Phi Theta Kappa Tuition Scholarship,** Aug. 2016.

Work Experience

- **Assistant Research Professor** in **Auburn University Transportation Research Institute (AUTRI)** Sep. 2023 - Present
  Samuel Ginn College of Engineering

- **IT Manager** for **Center for Applied Transportation Sciences (CATS)** Jul. 2022 – Aug. 2023
  Department of Civil and Architectural Engineering and Mechanics (CAEM), University of Arizona (UA)
  - Performed advanced database development and optimization, managed front end and back end IT infrastructure, developed Traffic Management Center wall, developed custom computing solutions, managed data streams, and developed unified web infrastructure. Led IT team, including undergraduate assistants, to assist center members with IT and data-related needs.

- **Graduate Research Assistant** in **Smart Transportation Lab** May. 2018 – Aug. 2023
  Department of Civil and Architectural Engineering and Mechanics (CAEM), University of Arizona (UA)
  - Developed advanced data-visualization platforms, developed backend server architecture, managed databases, performed advanced database development and optimization, conducted project meetings and had private meetings with project clients, wrote scope-of-work documents for projects, submitted journal papers, presented research findings at national/international conferences, used machine learning and data science techniques.

- **Undergraduate Research Assistant** in **Smart Transportation Lab** Jan. 2017 – May 2018
  Department of Civil and Architectural Engineering and Mechanics (CAEM), University of Arizona (UA)
  - Assisted with data collection, developed websites, developed data-visualization platforms, performed simple database development, assisted with journal paper proofreading and editing.

Research Grants

1. **Co-PI [25% Credit],** *A Data-Driven Comprehensive Action Plan for Multimodal Campus Road Safety*. USDOT Safe Streets and Roads for All (SS4A). Awarded Oct. 2026, project period pending. *$1.09 million*
2. **Co-PI [20% Credit],** *Public Transportation Management and Rural Transit Assistance Program*. Alabama Department of Transportation (ALDOT). Oct. 2026 ~ Sep. 2029. *$3.98 million*
3. **Co-PI (AUTRI Team – Proposal led and submitted by AUTRI),** *Research and Technology to Promote Safety of Commercial Motor Vehicles, Drivers, And Carriers: Indefinite Delivery, Indefinite Quantity (IDIQ) Proposal.* Federal Motor Carrier Safety Administration (FMCSA). Jul. 2026 ~ Jul. 2031. *One of twelve selected nationally to perform task orders totaling up to $60 million.*
4. **Co-PI [15% Credit],** *Safety and Operation Analysis for Rectangular Rapid Flashing Beacon Warrants*. USDOT Safe Streets and Roads for All (SS4A). Mar. 2026~Sep. 2028. *$1.05 million*
5. **Co-PI (AUTRI Team) [25% Credit],** *Rail Center for Research Enhancing Short-line Transportation (RailCREST).* Federal Railroad Administration (FRA). Aug. 2025~Jul. 2028. *$1.83 million*

6. **Research Team Leader (AUTRI Team)**, *Safe and Efficient Entry Control Points at Military Installations: Phase I*. Department of Defense (DOD). Aug. 2023 ~Present. *~$1.4 million*
7. **Lead Graduate Research Assistant,** *Traffic Data Analytics Platform Development for TSMO: Statewide Mobility Analytics in Real-Time (SMART) Tool Expansion*. Arizona Department of Transportation (ADOT). Apr. 2022~Aug. 2023. *~$200,000*
8. **Lead Graduate Research Assistant,** *Traffic Data Analytics Platform Development for TSMO – Expansion of Ramp Metering Evaluation Tool*. Arizona Department of Transportation (ADOT). Feb. 2020~Jan. 2022. *~$200,000*
9. **Lead Graduate Research Assistant,** *Data-Driven Evaluation for ADOT Ramp Metering*. Arizona Department of Transportation (ADOT). May 2018~Aug. 2019. *~$120,102*
10. **Research Assistant,** *Optimizing Traffic Signals Using Multi-Source Data – a Multi-Phase Project.* City of Tucson, Arizona. Aug. 2017~ Jul. 2018. *~$100,000*

Journal Publications

Published

1. Sushmita Bhandari, **Adrian Cottam**, Xi Zhang, Alyssa Ryan, and Yao-Jan Wu (2026). *Integrating Vehicle Probe Speed and Crash Data for Statewide Traffic and Road Safety Enhancement*. Transportation Research Record. https://doi.org/10.1177/03611981261425969
2. Xiaofeng Li, Hyunsooh Noh, **Adrian Cottam**, Qinzheng Wang (2026). *Leveraging Connected-Vehicle Data to Identify and Analyze Park Service Areas*. Cities. https://doi.org/10.1016/j.cities.2025.106665
3. Ernest Tufuor, **Adrian Cottam**, and Laurence Rilett (2025). *A Statewide Highway System Network Analysis: Impact of Probe Vehicle Composition.* Transportation Research Record. https://doi.org/10.1177/03611981251393235
4. Gabriel Gedaliah Geffen, **Adrian Cottam**, Henrick Haule, and Yao-Jan Wu (2025). *Evaluation of Factors Influencing Ramp Metering Compliance Using Controller Event-Based Data.* Transportation Research Record. https://doi.org/10.1177/03611981251347599
5. Abolfazl Karimpour, **Adrian Cottam**, Anthony Altieri, Ellwood Hanrahan II (2025). *Automated Statewide Estimation of Crash-Induced Delay and Queueing using Crowdsourced Data*. Case Studies on Transport Policy. https://doi.org/10.1016/j.cstp.2025.101565
6. **Adrian Cottam**, Xiaofeng Li, and Yao-Jan Wu (2025). *VEMLAN: Imputing Missing Data for Failed Freeway Traffic Sensors.* Journal of Intelligent Transportation Systems. https://doi.org/10.1080/15472450.2025.2477785
7. **Adrian Cottam**, Xiaofeng Li, and Yao-jan Wu (2024). *Machine-Learning Approach for Estimating Passenger Car Equivalent Factors using Crowdsourced Data.* Transportmetrica A: Transport Science. https://doi.org/10.1080/23249935.2024.2377600
8. **Adrian Cottam**, Xiaofeng Li, Xiaobo Ma, and Yao-jan Wu (2024). *Large-scale Freeway Traffic Volume Estimation using Crowdsourced Speed Data: A Case Study in Arizona*. Journal of Transportation Engineering, Part A: Systems. (https://doi.org/10.1061/JTEPBS/TEENG-8304)
9. **Adrian Cottam**, Xiaofeng Li, Mohammad Razaur Rahman Shaon, and Yao-jan Wu (2024). *Investigating the Impacts of E-Scooters on Bike-sharing System Ridership: A Data-Driven Study in Tucson, Arizona.* International Journal of Sustainable Transportation. (https://doi.org/10.1080/15568318.2023.2288122)
10. Xiaofeng Li, **Adrian Cottam**, and Yao-Jan Wu (2023). *Transit Arrival Time Prediction using Interaction Networks.* IEEE Transactions on Intelligent Transportation Systems. (https://doi.org/10.1109/TITS.2023.3238289)
11. Xiaoling Luo, **Adrian Cottam**, Yao-Jan Wu, and Yangsheng Jiang (2021). *Hybrid-Data Approach for Estimating Trip Purposes.* Transportation Research Record: Journal of the Transportation Research Board. (https://doi.org/10.1177/03611981211018474)
12. Xiaofeng Li, **Adrian Cottam**, Yao-Jan Wu, and Alireza Khani (2020). *Can a bikesharing system reduce fuel consumption? Case study in Tucson, Arizona*. Transportation Research Part D: Transport and Environment. (https://doi.org/10.1016/j.trd.2020.102604)
13. Xiaofeng Li, Alexander Weber, **Adrian Cottam**, and Yao-Jan Wu (2019). *Impacts of Changing from Permissive/Protected Left-Turn to Protected-Only Phasing. Case Study in the City of Tucson, Arizona*. Transportation Research Record: Journal of the Transportation Research Board. (https://doi.org/10.1177/0361198119842108)

Under Review

1. **Adrian Cottam**, Xiaofeng Li, and Yao-jan Wu. *Investigating Interactions between Sidewalk Autonomous Delivery Robots and Vehicular Traffic at Stop-Controlled Crosswalks.* *- Submitted to Transportation Research Part F, Pending Review.*

Pre-Print

1. Xiaobo Ma, **Adrian Cottam**, Mohammad Razaur Rahman Shaon, and Yao-Jan Wu (2023). *A Transfer Learning Framework for Proactive Ramp Metering Performance Assessment.* arXiv:2308.03542 (https://doi.org/10.48550/arXiv.2308.03542)

Conference Presentations

Peer-Reviewed Conference Papers

1. **Adrian Cottam**, Corey McDaniels, Griffin Herring, Ben Tuttle, and Laurence Rilett (2026). *AI Safety Analyst: Improving Accessibility to Data-Driven Transportation Safety Assessment with Multimodal Large Language Models*. Accepted for presentation at the 106th Annual Meeting of the Transportation Research Board in Washington, DC in January 2027.
2. **Adrian Cottam**, Amee Kivette, Ernest Tufuor, Corey McDaniels, and Laurence Rilett (2026). *Zero-Shot Open-Vocabulary Annotation of Roadside Assets Within Lidar Data Using Large Language Models*. Presented at the 105th Annual Meeting of the Transportation Research Board in Washington, DC.
3. Ernest Tufuor, **Adrian Cottam**, and Laurence Rilett (2025). *A Statewide Highway System Network Analysis: Impact of Probe Vehicle Composition*. Presented at the 104th Annual Meeting of the Transportation Research Board in Washington, DC. *Earned 2025 Best Overall Paper Award from the Committee on Highway Traffic Monitoring (ACP70)*
4. Gabriel Gedaliah Geffen, **Adrian Cottam**, Henrick Haule, and Yao-Jan Wu (2025). *Evaluation of Factors Influencing Ramp Metering Compliance Using Controller Event-Based Data*. Presented at the 104th Annual Meeting of the Transportation Research Board in Washington, DC. *Earned Young Professional / Student Best Paper Award from the Freeway Operations Committee (ACP20).*
5. Abolfazl Karimpour, Anthony Altieri, **Adrian Cottam**, Ellwood Hanrahan II (2025). *Automated Statewide Estimation of Crash-Induced Delay and Queueing using Crowdsourced Data*. Presented at the 104th Annual Meeting of the Transportation Research Board in Washington, DC.
6. Sushmita Bhandari, **Adrian Cottam**, Xi Zhang, Alyssa Ryan, Yao-Jan Wu (2025). *Integrating Crowdsourced Speed Data and Crash Data for Statewide Traffic Management and Road Safety Enhancement*. Presented at the 104th Annual Meeting of the Transportation Research Board in Washington, DC.
7. Xiaofeng Li, Hyunsoo Noh, **Adrian Cottam**, Qinzheng Wang (2025). *Leveraging Connected-Vehicle Data to Identify Park Service Areas for Driving-Only Users*. Presented at the 104th Annual Meeting of the Transportation Research Board in Washington, DC.
8. **Adrian Cottam**, Xiaofeng Li, and Yao-Jan Wu (2024). *Machine-Learning Approach for Estimating Passenger Car Equivalent Factors using Crowdsourced Data.* Presented at the 103rd Annual Meeting of the Transportation Research Board in Washington, DC.
9. **Adrian Cottam**, Xiaofeng Li, Cynthia Navarro, and Yao-Jan Wu (2024). *Investigating Interactions between Sidewalk Autonomous Delivery Robots and Vehicular Traffic at Stop-Controlled Crosswalks.* Presented at the 103rd Annual Meeting of the Transportation Research Board in Washington, DC.
10. Xiaobo Ma, **Adrian Cottam**, Mohammad Razaur Rahman Shaon, and Yao-Jan Wu (2024). *Application of Transfer Learning for Proactive Ramp Metering Performance Assessment.* Presented at the 103rd Annual Meeting of the Transportation Research Board in Washington, DC.
11. **Adrian Cottam**, Xiaofeng Li, and Yao-Jan Wu (2023). *Machine-Learning Approach for Estimating Passenger Car Equivalency Factors using Crowdsourced Data.* Presented at TRB Committee on Highway Capacity and Quality of Service (ACP40) Uninterrupted Flow Subcommittee Quarterly Meeting.
12. **Adrian Cottam**, Xiaofeng Li, and Yao-Jan Wu (2023). *VEMLAN: Imputing Missing Data for Failed Freeway Traffic Sensors.* Presented at the 102nd Annual Meeting of the Transportation Research Board in Washington, DC.
13. Xiaofeng Li, **Adrian Cottam**, and Yao-jan Wu (2022). *Transit Arrival Time Prediction by Fusing GTFS and*

Crowdsourced Data. Presented at the 101st Annual Meeting of the Transportation Research Board in Washington, DC.

14. **Adrian Cottam**, Xiaobo Ma, Xiaofeng Li, and Yao-jan Wu (2021). *Large-scale Freeway Traffic Volume Estimation using Crowdsourced Speed Data: A Case Study in Arizona.* Presented at the 100th Annual Meeting of the Transportation Research Board in Washington, DC.
15. **Adrian Cottam**, Xiaofeng Li, Mohammad Razaur Rahman Shaon, and Yao-jan Wu (2021). *Investigating the Impacts of E-Scooters on Bike-sharing System Ridership: A Data-Driven Study in Tucson, Arizona.* Presented at the 100th Annual Meeting of the Transportation Research Board in Washington, DC.
16. Yang Liang, Xiaoqing Zeng, Xiaofeng Li, **Adrian Cottam**, and Yao-Jan Wu (2020). *Quantifying Mobility Impact of Pedestrian Hybrid Beacon Signal on Arterial Traffic: Case Study in Tucson, Arizona*. Presented at the 99th Annual Meeting of the Transportation Research Board in Washington, DC.
17. Xiaofeng Li, Alexander Weber, **Adrian Cottam**, and Yao-Jan Wu (2019). *Impacts of Changing from Permissive/Protected Left-Turn to Protected-Only Phasing. Case Study in the City of Tucson, Arizona*. Presented at the 98th Annual Meeting of the Transportation Research Board in Washington, DC.

Professional Presentations

1. **Adrian Cottam**, Amee Kivette, Ernest Tufuor, Corey McDaniels, and Laurence Rilett (2026). *Zero-Shot Open-Vocabulary Annotation of Roadside Assets Within Lidar Data Using Large Language Models*. Presented at the 2026 ASCE International Conference on Transportation & Development in Detroit, MI.
2. Corey McDaniels, Timo Mueller, **Adrian Cottam**, Ernest Tufuor, and Laurence Rilett (2026). *A Scalable Automated LiDAR-Based Roadway Digital Twin Creation Framework for Virtual Curve Safety Studies*. Presented at the 2026 ASCE International Conference on Transportation & Development in Detroit, MI.
3. **Adrian Cottam**, Corey McDaniels, Laurence Rilett, and Steven Carter (2025). *Stochastic Calibration of Entry Control Facility Microsimulation Models Using Distribution Matching*. Presented at the 2025 ASCE International Conference on Transportation & Development in Glendale, AZ.
4. Ernest Tufuor, **Adrian Cottam**, and Laurence Rilett (2025). *Impact Of Probe Vehicle Data and Roadway Features on Travel Time Reliability: A Statewide Analysis*. Presented at the 2025 ASCE International Conference on Transportation & Development in Glendale, AZ.
5. **Adrian Cottam** (2024). *Impacts of Probe Vehicle Compositions on NPMRDS.* Presented at the 2024 GAITE/ALSITE Technical Exchange in Auburn, AL.
6. **Adrian Cottam**, Xiaofeng Li, and Yao-jan Wu (2024). *Machine-Learning Approach for Estimating Passenger Car Equivalent Factors using Crowdsourced Data.* Presented at the 2024 ASCE International Conference on Transportation & Development in Atlanta, GA.
7. John Roberts, **Adrian Cottam** (2022). *Beyond Rush Hour: Using Ramp Metering and Event-Based Data for Incident Response.* Presented at the 29th ITS Arizona Annual Conference in Mesa, AZ.
8. **Adrian Cottam**, Xiaobo Ma, Yong Ge, and Yao-jan Wu (2022). *Spatial Transferability of Volume Estimation Models: A Multi-Model Analysis with Spatial Embedding.* Presented at the 101st Annual Meeting of the Transportation Research Board Eisenhower Scholars Poster Session in Washington, DC.
9. Steven Cheshko, **Adrian Cottam** (2021). *Adapting to a Pandemic: Ramp Metering with Changing Traffic Patterns*. Presented at the 28th ITS Arizona Annual Conference in Mesa, AZ.
10. **Adrian Cottam** (2018). *Effects of Street Lighting on the Severity of Vulnerable Road User Crashes*. Presented at the 2018 COTA-ISETT conference in Honolulu, HI.

Teaching Experience

Instructor

1. Spring 2025 (Auburn University): CIVL 5500/6500 Traffic Engineering Analysis **(Instructor)**. [Credits: 3, Evaluation: 5.3/6, Enrollment 20]
2. Fall 2022 (University of Arizona): CE 363 Transport Engineering and Pavement Design **(Instructor)**. [Credits: 4, Evaluation: 9.6/10, Enrollment 28]
3. Spring 2022 (University of Arizona): CE 465/565 Transportation Data Management and Analysis **(Technical Session Instructor)**. [Credits: 3, Evaluation: 5/5, Enrollment: 7]

Teaching Assistant

4. Fall 2018 (University of Arizona): CE 363 Transport Engineering and Pavement Design (Teaching Assistant). [Credits: 4, Evaluation: 4.3/5, Enrollment: 39]
5. Fall 2017 (University of Arizona): CE 363 Transport Engineering and Pavement Design (Teaching Assistant). [Credits: 4, Evaluation: 4.5/5, Enrollment: 51]

Technical Reports

1. Yao-Jan Wu and **Adrian Cottam**. Traffic Data Analytics Platform Development for TSMO – Expansion of Ramp Metering Evaluation Tool, Arizona Department of Transportation (ADOT). Mar. 2022.
2. Yao-Jan Wu, **Adrian Cottam**, and Xiaobo Ma. Data-Driven Evaluation for ADOT Ramp Metering: Developing Ramp Metering Evaluation Tool, Arizona Department of Transportation (ADOT). Sep. 2019.

Public Services

Paper Review

1. Transportation Research Record, 2024. Provided 2 article reviews.
2. Journal of Transportation Research Part A, 2023-2024. Provided 2 article reviews.
3. ASCE Journal of Transportation Engineering, Part A: Systems, 2023-2024. Provided 5 article reviews.
4. Transportation Research Board Annual Meeting, 2021 – 2023. Provided 10 article reviews.

Committee Membership

1. 2025: Member of Transportation Research Board Standing Committee on Intelligent Transportation Systems (ACP15).
2. 2024-2025: Paper Review Coordinator for Transportation Research Board Standing Committee on Intelligent Transportation Systems (ACP15).
3. 2023-2025: Friend of Transportation Research Board Standing Committee on Intelligent Transportation Systems (ACP15). Provided committee notetaking services.
4. 2019-Present: Friend of Transportation Research Board Standing Committee on Artificial Intelligence and Advanced Computing Applications (AED50). Provided professional reviewer services.
5. 2019-Present: Friend of Transportation Research Board Standing Committee on Information Systems and Technology (AED30). Provided professional reviewer services.

Student Organizations

1. AY 2020-2021: University of Arizona Institute of Transportation Engineers (UAITE) chapter president.
2. AY 2019-2020: University of Arizona Institute of Transportation Engineers (UAITE) chapter treasurer.
3. AY 2018-2019: University of Arizona Institute of Transportation Engineers (UAITE) chapter vice-president.

Mentorship

1. 2019-2021: Mentored Josh Swain (Undergraduate) to perform data analysis, server setup, and introductory systems development.
2. 2020-2022: Mentored Dylan Molina (Undergraduate) to write R code to develop web applications and set up backend systems.
3. 2021-2023: Mentored Gabriel Geffen (Undergraduate) to write R code, develop web applications including backend, handle IT related tasks, and submit a first author paper for TRB.
4. 2021-2023: Mentored Xi Zhang (Graduate) to develop containerized traffic data analysis web applications in R using arterial data for multiple local agencies.
5. 2021-2023: Mentored Pramesh Pudasaini (Graduate) to develop containerized traffic data analysis web applications in R using spatial data merging.
6. 2021-2023: Mentored Pouya Jalali Khalilabadi (Graduate) to develop containerized traffic data analysis web applications in R to provide arterial traffic and safety analyses.
7. 2023-Present: Mentored Corey McDaniels to develop a master's thesis topic evaluating Entry Control Facilities (ECFs) threat vehicle scenarios using reinforcement learning.

8. 2024-Present: Mentored Kevien Shelton to perform research creating representations of Entry Control Facilities (ECFs) using geometric data and microsimulation software.
9. 2024-Present: Mentored Griffin Herring to perform research supporting his master's thesis evaluating imitation learning for autonomous driving applications.
10. 2024-2026: Mentored Amee Kivette to perform research supporting her master's thesis developing a driving simulator and corresponding remote control web application, as well as developing a CMV speeding evaluation web application.
11. 2025-2026: Mentored Garrett Scott to perform research supporting his master's thesis evaluating imitation learning for autonomous driving. Served as committee member for thesis.
12. 2025-2026: Mentored Keanna Tennyson to perform research supporting her master's thesis developing a large language model chatbot for speed data dashboard navigation using natural language.
13. 2025-Present: Mentored Ben Tuttle to perform research supporting his master's thesis developing stochastic methods of threat vehicle travel times at military installations. Served as committee member for thesis.
14. 2025-Present: Mentored Timo Mueller to perform research supporting his doctoral dissertation developing low-cost highway rail grade crossing safety monitoring devices.